

Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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2026-03-18

Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить перенаправление ввода-вывода
- 2 Изучить работу фильтров
- 3 Изучить команду поиска
- 4 Ознакомиться с управлением процессами
- 5 Ознакомиться с командами `df` `du`

2. Процесс выполнения лабораторной работы

```
jrajabov@jradjabov:~$ ls /etc/ > file.txt
jrajabov@jradjabov:~$ ls >> file.txt
jrajabov@jradjabov:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
```

Рисунок 1: Запись в файл

```
jradjabov@jradjabov:~$ grep .conf file.txt > conf.txt
jradjabov@jradjabov:~$ cat conf.txt
anthy-unicode.conf
asound.conf
brltty.conf
chrony.conf
cpupower-service.conf
dconf
dnsmasq.conf
dracut.conf
dracut.conf.d
fprintd.conf
fuse.conf
host.conf
idmapd.conf
kdump.conf
krb5.conf
krb5.conf.d
ld.so.conf
ld.so.conf.d
libaudit.conf
libuser.conf
```

Рисунок 2: Поиск расширения .conf


```
ro/project-personal/stage02/report/_resources/csl
/home/jrajabov/work/study/2025-2026/Операционные системы/2026-1--study--os-int
ro/project-personal/stage03/report/_resources/csl
/home/jrajabov/work/study/2025-2026/Операционные системы/2026-1--study--os-int
ro/project-personal/stage03/report/bib/cite.bib
/home/jrajabov/work/study/2025-2026/Операционные системы/2026-1--study--os-int
ro/project-personal/stage04/report/_resources/csl
/home/jrajabov/work/study/2025-2026/Операционные системы/2026-1--study--os-int
ro/project-personal/stage04/report/bib/cite.bib
/home/jrajabov/work/study/2025-2026/Операционные системы/2026-1--study--os-int
ro/project-personal/stage05/report/_resources/csl
/home/jrajabov/work/study/2025-2026/Операционные системы/2026-1--study--os-int
ro/project-personal/stage05/report/bib/cite.bib
/home/jrajabov/work/study/2025-2026/Операционные системы/2026-1--study--os-int
ro/project-personal/stage06/report/_resources/csl
/home/jrajabov/work/study/2025-2026/Операционные системы/2026-1--study--os-int
ro/project-personal/stage06/report/bib/cite.bib
/home/jrajabov/work/study/2025-2026/Операционные системы/2026-1--study--os-int
ro/presentation/report/_resources/csl
/home/jrajabov/work/study/2025-2026/Операционные системы/2026-1--study--os-int
ro/presentation/report/bib/cite.bib
/home/jrajabov/git-extended/.git/hooks/commit-msg.sample
/home/jrajabov/git-extended/.git/objects/cf
/home/jrajabov/git-extended/.git/objects/1f/c8552865c2c2bbad7a5c4d93684af76165
8c11
/home/jrajabov/git-extended/.git/config
/home/jrajabov/conf.txt
jrajabov@jradjabov:~$
```

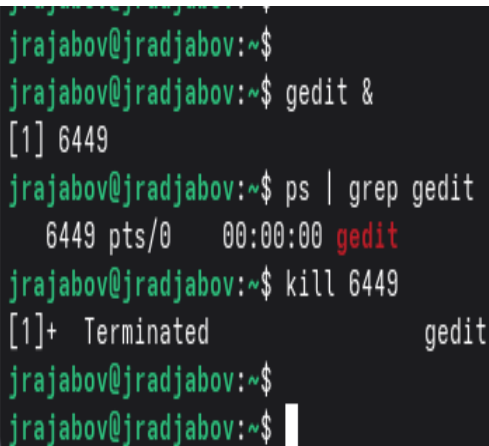
Рисунок 3: Поиск файлов

```
/etc/foomatic/hashes.d/hashes.new
find: '/etc/libvirt': Permission denied
/etc/hp
/etc/hp/hplip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
find: '/etc/lvm/archive': Permission denied
/etc/logrotate.d/httpd
find: '/etc/lvm/backup': Permission denied
find: '/etc/lvm/cache': Permission denied
find: '/etc/lvm/devices': Permission denied
find: '/etc/nftables': Permission denied
find: '/etc/openvpn/client': Permission denied
find: '/etc/openvpn/server': Permission denied
/etc/nvme/hostnqn
/etc/nvme/hostid
find: '/etc/pki/rsyslog': Permission denied
find: '/etc/polkit-1/localauthority': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/sos/cleaner': Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
/etc/sane.d/hp5400.conf
:find: '/etc/ssh/sshd_config.d': Permission denied
find: '/etc/sss': Permission denied
find: '/etc/sudoers.d': Permission denied
find: '/etc/userdb': Permission denied
```

Рисунок 4: Поиск файлов

```
jrajabov@jradjabov:~$  
jrajabov@jradjabov:~$ find ~ -name "log*" > logfile &  
[1] 6412  
jrajabov@jradjabov:~$  
[1]+  Done                  find ~ -name "log*" > logfile  
jrajabov@jradjabov:~$ rm logfile  
jrajabov@jradjabov:~$
```

Рисунок 5: Фоновый запуск процесса



```
jrajabov@jradjabov:~$  
jrajabov@jradjabov:~$ gedit &  
[1] 6449  
jrajabov@jradjabov:~$ ps | grep gedit  
    6449 pts/0    00:00:00 gedit  
jrajabov@jradjabov:~$ kill 6449  
[1]+  Terminated                  gedit  
jrajabov@jradjabov:~$  
jrajabov@jradjabov:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

```
DF(1)                                User Commands                                DF(1)

NAME
    df - report file system space usage

SYNOPSIS
    df [OPTION]... [FILE]...

DESCRIPTION
    This manual page documents the GNU version of df.  df displays the
    amount of space available on the file system containing each file name
    argument.  If no file name is given, the space available on all cur-
    rently mounted file systems is shown.  Space is shown in 1K blocks by
    default, unless the environment variable POSIXLY_CORRECT is set, in
    which case 512-byte blocks are used.

    If an argument is the absolute file name of a device node containing a
    mounted file system, df shows the space available on that file system
    rather than on the file system containing the device node.  This ver-
    sion of df cannot show the space available on unmounted file systems,
    because on most kinds of systems doing so requires non-portable inti-
    mate knowledge of file system structures.

OPTIONS
    Show information about the file system on which each FILE resides, or
    all file systems by default.

    Mandatory arguments to long options are mandatory for short options
    too.

    -a, --all
    Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

```
DU(1)                                User Commands                                DU(1)

NAME
    du - estimate file space usage

SYNOPSIS
    du [OPTION]... [FILE]...
    du [OPTION]... --files0-from=F

DESCRIPTION
    Summarize device usage of the set of FILEs, recursively for directories.

    Mandatory arguments to long options are mandatory for short options too.

    -0, --null
        end each output line with NUL, not newline

    -a, --all
        write counts for all files, not just directories

    --apparent-size
        print apparent sizes rather than device usage; although the apparent size is usually smaller, it may be larger due to holes in ('sparse') files, internal fragmentation, indirect blocks, and the like

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576 bytes; see SIZE format below

Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

```
jrajabov@jradjabov:~$  
jrajabov@jradjabov:~$ df  
Filesystem      1K-blocks      Used Available Use% Mounted on  
/dev/nvme0n1p3 155186176 89533968 64981856 58% /  
devtmpfs        4011268      0      4011268  0% /dev  
tmpfs           4035644       92    4035552  1% /dev/shm  
tmpfs           1614260      1952    1612308  1% /run  
tmpfs           1024          0        1024  0% /run/credentials/systemd-jour  
nald.service  
tmpfs           4035644      1612    4034032  1% /tmp  
/dev/nvme0n1p3 155186176 89533968 64981856 58% /home  
/dev/nvme0n1p2 1992552      518800    1352512 28% /boot  
/dev/loop2      49280        49280          0 100% /var/lib/snapd/snap/snapd/259  
35  
/dev/loop0      68480        68480          0 100% /var/lib/snapd/snap/core24/13  
49  
/dev/loop1      115712       115712          0 100% /var/lib/snapd/snap/hugo/2567  
6  
tmpfs           1024          0        1024  0% /run/credentials/systemd-reso  
lved.service  
tmpfs           807128       168      806960  1% /run/user/1025  
/dev/sr0        2677920     2677920          0 100% /run/media/jrajabov/Fedora-WS  
-Live-43  
jrajabov@jradjabov:~$
```

Рисунок 9: Справка по команде du

```
4      ./git-extended/.git/objects/52
4      ./git-extended/.git/objects/46
4      ./git-extended/.git/objects/6b
4      ./git-extended/.git/objects/79
4      ./git-extended/.git/objects/e2
4      ./git-extended/.git/objects/7f
4      ./git-extended/.git/objects/2e
76     ./git-extended/.git/objects
8      ./git-extended/.git/logs/refs/heads
8      ./git-extended/.git/logs/refs/remotes/origin
8      ./git-extended/.git/logs/refs/remotes
16     ./git-extended/.git/logs/refs
20     ./git-extended/.git/logs
212    ./git-extended/.git
220    ./git-extended
0      ./monthly
0      ./reports/monthly/monthly
0      ./reports/monthly
0      ./reports
4      ./ski.plases/equipment
0      ./ski.plases/plans
4      ./ski.plases
0      ./australia
0      ./play/games/play
0      ./play/games
0      ./play
818780 .
jrajabov@jradjabov:~$
```

Рисунок 10: Запуск команды du


```
/home/jrajabov/git-extended/.git/objects/01  
/home/jrajabov/git-extended/.git/objects/1d  
/home/jrajabov/git-extended/.git/objects/1f  
/home/jrajabov/git-extended/.git/objects/15  
/home/jrajabov/git-extended/.git/objects/82  
/home/jrajabov/git-extended/.git/objects/32  
/home/jrajabov/git-extended/.git/objects/46  
/home/jrajabov/git-extended/.git/objects/6b  
/home/jrajabov/git-extended/.git/objects/79  
/home/jrajabov/git-extended/.git/objects/e2  
/home/jrajabov/git-extended/.git/objects/7f  
/home/jrajabov/git-extended/.git/objects/2e  
/home/jrajabov/git-extended/.git/logs  
/home/jrajabov/git-extended/.git/logs/refs  
/home/jrajabov/git-extended/.git/logs/refs/heads  
/home/jrajabov/git-extended/.git/logs/refs/remotes  
/home/jrajabov/git-extended/.git/logs/refs/remotes/origin  
/home/jrajabov/monthly  
/home/jrajabov/reports  
/home/jrajabov/reports/monthly  
/home/jrajabov/reports/monthly/monthly  
/home/jrajabov/ski.plases  
/home/jrajabov/ski.plases/equipment  
/home/jrajabov/ski.plases/plans  
/home/jrajabov/australia  
/home/jrajabov/play  
/home/jrajabov/play/games  
/home/jrajabov/play/games/play  
jrajabov@jradjabov:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.